

Asma Mam — Exam Answer Sheet

Society, Environment & Professional Ethics | 10-Mark Questions

Question 1 (10 Marks)

Q: What are the Sustainable Development Goals (SDGs)? Discuss.

Introduction:

The Sustainable Development Goals (SDGs), also known as the Global Goals, are a set of 17 interconnected goals adopted by all 193 United Nations Member States in September 2015 at the UN Sustainable Development Summit in New York. They form the core of the **2030**

Agenda for Sustainable Development — a universal call to action to end poverty, protect the planet, and ensure peace and prosperity for all people by the year 2030. The SDGs replaced the earlier Millennium Development Goals (MDGs) and are broader in scope, addressing the root causes of poverty and inequality.

Key Features of the SDGs:

- 17 Goals with **169 associated targets**
- Applicable to **all countries** — both developed and developing
- Built around the **5 Ps: People, Planet, Prosperity, Peace, and Partnership**
- Not legally binding, but countries must develop national frameworks for implementation
- Progress is monitored through a global indicator framework

The 17 Sustainable Development Goals:

No.	Goal	No.	Goal
1	No Poverty	10	Reduced Inequalities
2	Zero Hunger	11	Sustainable Cities & Communities
3	Good Health & Well-being	12	Responsible Consumption & Production
4	Quality Education	13	Climate Action
5	Gender Equality	14	Life Below Water
6	Clean Water & Sanitation	15	Life on Land
7	Affordable & Clean Energy	16	Peace, Justice & Strong Institutions
8	Decent Work & Economic Growth	17	Partnerships for the Goals
9	Industry, Innovation & Infrastructure		

Importance of SDGs:

The SDGs recognize that social, economic, and environmental sustainability are deeply interconnected — progress in one area affects outcomes in others. They ensure that **no one is left behind**, giving special attention to the needs of the poorest and most vulnerable

populations. Governments, civil society, the private sector, and individuals all share responsibility for achieving these goals.

Conclusion:

The SDGs represent humanity's most comprehensive commitment to building a better and more sustainable future. Achieving all 17 goals by 2030 requires strong political will, international cooperation, sufficient financing, and the active participation of all stakeholders across the globe.

Question 2 (10 Marks)

Q: What is Intellectual Property? Explain its types.

Introduction:

Intellectual Property (IP) refers to creations of the human mind — such as inventions, literary and artistic works, designs, symbols, names, and images used in commerce. Just like physical property, intellectual property is an asset that belongs to its creator. IP is protected by law through mechanisms such as patents, copyrights, trademarks, and trade secrets, which enable people to earn recognition and financial benefit from what they invent or create. The IP system aims to foster an environment where creativity and innovation can flourish.

Types of Intellectual Property:

1. Copyright

Copyright protects original literary and artistic works such as books, music, films, photographs, paintings, computer software, and databases. Copyright arises automatically upon the creation of a work — no registration is required. The creator gets exclusive rights to reproduce, distribute, perform, and display the work. Duration is generally the life of the author plus 70 years.

Example: Books, songs, films, software

2. Patent

A patent protects new inventions and grants the inventor the exclusive right to make, use, or sell the invention for a limited period — generally 20 years. To receive a patent, the invention must be new, useful, and non-obvious. In exchange for this exclusive right, the inventor must publicly disclose how the invention works. Types include utility patents, design patents, and plant patents.

Example: New machines, medicines, processes

3. Trademark

A trademark is a word, phrase, symbol, or design that identifies and distinguishes the source of goods or services of one party from those of others. Unlike patents, trademarks can last

indefinitely as long as they continue to be used. Trademarks protect brand identity and prevent consumer confusion in the marketplace.

Example: Nike logo, Coca-Cola name, slogans

4. Trade Secret

A trade secret is confidential business information that provides a competitive advantage — such as formulas, processes, customer lists, or internal strategies. Unlike other forms of IP, trade secrets cannot be registered. They are protected by keeping them secret through confidentiality agreements and restricted access. They have no expiration date as long as secrecy is maintained.

Example: Coca-Cola recipe, business processes

Importance of Intellectual Property:

IP protection encourages innovation and creativity by ensuring that creators and inventors can benefit from their work. It promotes economic growth, supports fair competition, and protects consumers. In computing, IP is especially significant — software, algorithms, and digital content all fall under various IP protections. The ACM Code of Ethics (Section 1.5) also requires computing professionals to respect intellectual property rights.

Conclusion:

Intellectual property is a vital legal framework that balances the rights of creators with the broader public interest. By protecting inventions, creative works, brand identities, and business secrets, IP law drives innovation and ensures a fair and competitive environment for all.

Question 3 (10 Marks)

Q: What is Digital Bangladesh? Discuss its objectives and how it evolved into Smart Bangladesh.

Introduction:

Digital Bangladesh is a visionary initiative of the Government of Bangladesh, introduced as part of the electoral manifesto of the Awami League in 2008. Officially launched in 2009, the vision — formally known as **Digital Bangladesh Vision 2021** — aimed to transform Bangladesh into a technologically developed, knowledge-based society by the year 2021. The core focus was on using Information and Communication Technology (ICT) to improve governance, promote economic growth, and enhance the quality of life for all citizens.

Objectives of Digital Bangladesh:

- Improve government service delivery using ICT
- Develop human capital and move toward a knowledge-based economy
- Diversify exports toward knowledge and IT-based products

- Ensure internet and digital connectivity for all citizens, especially in rural areas
- Promote e-commerce and digital financial services
- Reduce corruption through transparent, digitized governance

The 4 Pillars of Digital Bangladesh:

1. Digital Governance: The government focused on digitizing all major public services. Over 2,000 government services were automated and made available online. The goal was paperless offices and transparent administration through e-governance.

2. Human Capital Development: Bangladesh invested heavily in ICT education and digital literacy programs to prepare citizens for the digital economy. Vocational training and online education were promoted to build a skilled workforce.

3. IT Industry Promotion: The government encouraged the growth of the local IT and software industry to increase exports of digital products and services. Hi-Tech Parks and special economic zones were established to attract investment.

4. Connecting Citizens: Rural and underserved populations were connected through expanded mobile networks, internet infrastructure, and Union Digital Centers (UDCs), ensuring that the benefits of digitization reached every citizen.

Achievements of Digital Bangladesh:

Within a decade, Digital Bangladesh moved from dream to reality. Mobile phone penetration increased dramatically, internet connectivity expanded across the country, government services became accessible online, and Bangladesh began exporting ICT services globally. The initiative became a model for developing countries.

From Digital Bangladesh to Smart Bangladesh:

Building on the success of Digital Bangladesh Vision 2021, the government announced **Smart Bangladesh Vision 2041** on December 12, 2022. While Digital Bangladesh focused on digitizing services and connecting citizens, Smart Bangladesh goes further — using advanced technologies such as Artificial Intelligence (AI), Internet of Things (IoT), blockchain, big data, and robotics to transform every aspect of society.

The 4 Pillars of Smart Bangladesh:

1. Smart Citizen: Empowering every citizen with a 'digital first' mindset through widespread digital literacy programs.

2. Smart Government: 100% paperless government offices with hyper-personalized service platforms across healthcare, education, and agriculture.

3. Smart Society: A society built on digital ethics and values, including a cashless payment ecosystem, Smart Cities, and Smart Grid infrastructure.

4. Smart Economy: Making ICT one of Bangladesh's primary revenue-generating sectors by 2041 through a thriving digital economy and fintech industry.

Conclusion:

Digital Bangladesh was a landmark achievement that laid the foundation for a modern, connected nation. Smart Bangladesh Vision 2041 is the next bold step — aiming to make Bangladesh a high-income, developed country by 2041 by harnessing the full power of 21st century technologies, fulfilling the dream of a prosperous Golden Bangladesh.

Good luck in your exam! — Prepared from Asma Mam's syllabus